

11. Show that if $a < b$ then $-b < -a$.

$$(-a) - (-b) \in \mathbb{R}^+$$

$$(-a) - (-b) = -a + b = b - a$$

$$b - a \in \mathbb{R}^+$$

$$a < b$$

12. Show that if $0 < a < b$ then $\frac{1}{a} > \frac{1}{b}$

if $x > 0$ then $\frac{1}{x} > 0$

since $x \cdot \frac{1}{x} = 1 \neq 0$

so $\frac{1}{x} \neq 0$

$$a > 0, b > 0$$

closure $a \cdot b > 0$

so $\frac{1}{ab} > 0$ by the definition $ab \neq 0$

$$\frac{1}{a} - \frac{1}{b} = \frac{b-a}{ab} = (b-a) \cdot \frac{1}{ab}$$

$$(b-a) \cdot \frac{1}{ab} \in \mathbb{R}^+$$

$$\frac{1}{a} > \frac{1}{b}$$

13. Show that for all real numbers a, b, c and d :

(a) If $a < b$ and $c \leq d$ then $a + c < b + d$

$$(b+d) - (a+c) = (b-a) + (d-c) \quad (b+d) - (a+c) \in \mathbb{R}^+$$

$$b-a \in \mathbb{R}^+ \text{ and } d-c \in \mathbb{R}^+ \cup \{0\}$$

if $d-c \in \mathbb{R}^+$ the ^{difference} ~~sum~~ is positive by closure

if $d-c = 0$ the ~~sum~~ ^{difference} equals $b-a \in \mathbb{R}^+$

in both cases it is positive

(b) If $0 < a < b$ and $0 < c \leq d$ then $ac < bd$

$$bd - ac = b(d-c) + (b-a)c$$

since $b > 0$ and $d-c \in \mathbb{R}^+ \cup \{0\}$

$$b(d-c) \in \mathbb{R}^+ \cup \{0\}$$

$$b-a \in \mathbb{R}^+ \text{ and } c \in \mathbb{R}^+$$

$$\text{so, } (b-a)c \in \mathbb{R}^+$$

write
up
problem 16

$$ac - bc$$

$$c(a-b)$$

$$a \cdot c - b \cdot c$$

$$c(a-b) \dots$$

$$-c(-a+b) = -c(b-a)$$

consider

$$(b+d) - (a+c)$$

using the field axioms

$$(b+d) - (a+c) = (b-a) + (d-c)$$